Page No.:
Date: / /

UNIT-05 (MATHS-04)

T-test :-

Test-01 To test the Significance of the mean of a random Sample.

Working-Rule :-

1. Null Hypothesis H_0 :- $\mu = \mu_0$ (given) there is no significant diff^n b/w Sample mean and hypothetical Population mean.

2. Alternative Hypothesis $\mu \neq \mu_0$

3. $t = \dfrac{\bar{x} - \mu}{\dfrac{S}{\sqrt{n}}}$

$\bar{x}$ = Sample mean
n = Size of Sample
S = Standard Deviation

$$S = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$$

4. From t-test table Find Value of t_α
DOF = $(n-1)$

5. Conclusion:
→ If $|t| < t_\alpha$ than H_0 is accepted.
→ If $|t| > t_\alpha$ than H_0 is rejected.

Page No.
Date / /

Qu-01 The 9 items of a sample have the following values :- 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these values differ significantly from the assumed mean 47.5 ? [The tabulated values of $t_{0.05} = 2.31$ for 8 d.f.]

[AKTU-2021-22]

Sol^n

$n = 9$, $\mu = 47.5$

$t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$ $\qquad s^2 = \dfrac{\Sigma(x-\bar{x})^2}{n-1}$

$\bar{x}$ = mean of sample

μ = Population mean

S.D = Standard deviation of Population

n = Sample mean

x	$x - \bar{x}$ $x - 49.1$	$(x-\bar{x})^2$
45	-4.1	16.81
47	-2.1	4.41
50	.9	.81
52	2.9	8.41
48	-1.1	1.21
47	-2.1	4.41
49	-0.1	.01
53	3.9	15.21
51	1.9	3.61

$\Sigma(x-\bar{x})^2 = 54.89$

$\Sigma x = 442$

$\bar{x} = \dfrac{\Sigma x}{n} = \dfrac{442}{9} = 49.1$

Page No.:
Date: / /

$$S^2 = \frac{\Sigma(x-\bar{x})^2}{n-1} = \frac{54.89}{8} = 6.89 \qquad \boxed{S = 2.619}$$

$$t = \frac{\bar{x}-\mu}{\frac{S}{\sqrt{n}}} = \frac{49.1 - 47.5}{\frac{2.619}{\sqrt{9}}} \qquad \boxed{t = 1.8327}$$

$t_{cal} = 1.8327$ ~~t_{cal}~~

$t_{tab} = 2.31$ $t_{tab} > t_{cal}$

H_0 is accepted $|t| > t_\alpha$

Qu-2 Following is the data of defectives of 10 samples of size 100 each

Sample no	1	2	3	4	5	6	7	8	9	10
No of defn	15	11	9	6	5	4	3	2	7	1

Construct p-chart & state whether the Process is Statistical control?

[AKTU - 2021-22 / 2020-21]

Soln A control chart based on Fraction defective $\bar{p}$

$$\bar{p} = \frac{\text{Total no of defectives}}{\text{Total no of items inspected}}$$

$$\bar{p} = \frac{63}{10 \times 100} = \frac{63}{1000} = \boxed{0.63 = \bar{p}}$$

$$\boxed{\text{Control limit} = \bar{p} = 0.63}$$

Page No.
Date / /

$UCL = \bar{p} + 3\sigma$

$= \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$

$= 0.63 + 3\sqrt{\frac{0.63(1-0.63)}{100}}$

$\boxed{UCL = .13588}$

$LCL = \bar{p} - 3\sigma$

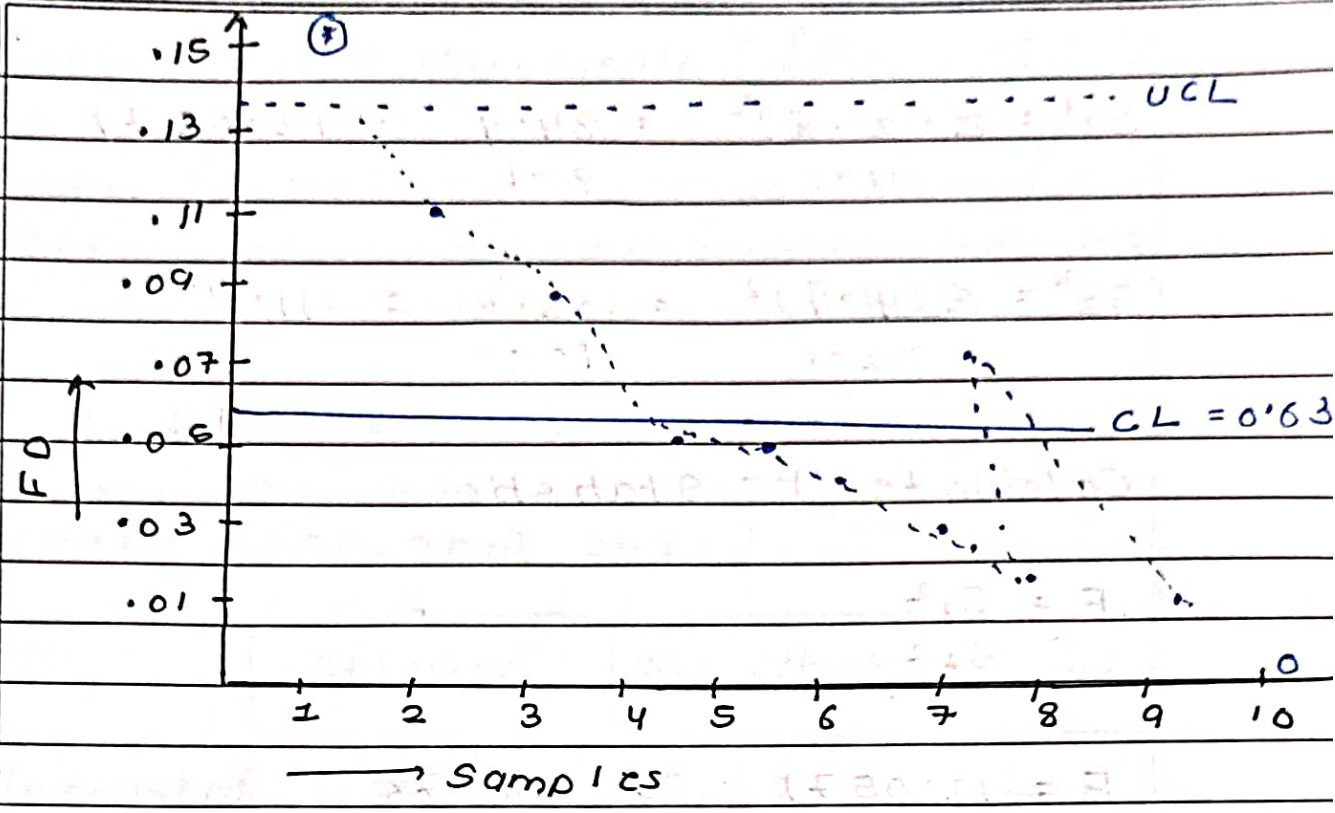
$= \bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$

$LCL = .063 - 3\sqrt{\frac{0.63(1-0.63)}{100}}$

$\boxed{LCL = -0.0098 \approx 0}$

Sample	Defective	Fraction Defective
1	15	15/100 = .15
2	11	11/100 = .11
3	9	9/100 = .09
4	6	6/100 = .06
5	5	5/100 = .05
6	4	4/100 = .04
7	3	3/100 = .03
8	2	2/100 = .02
9	7	7/100 = .07
10	1	1/100 = .01

Page No.:
Date: / /



Hence Process is not Under control Ans

Qu-03 In Two independent Sample Size 8 and 10, the sum of square of deviation of Sample Values From respective mean are 84.4 and 102.6. Test whether the diff^m of Variances of Population is Segment or not. Use 5% of level of significance [$F_{0.5}(7,9) = 3.29$]

[AKTU - 2021-22]

Sol^n Given data

- $n_1 = 8$, $S_1^2 = 84.4$
- $n_2 = 10$, $S_2^2 = 102.6$
- Critical Value $F_{0.05}(7, 9) = 3.29$

- Null Hypothesis H_0 $\sigma_1^2 = \sigma_2^2$
- Alternative Hypothesis $H_1 = \sigma_1^2 \neq \sigma_2^2$

Page No..
Date / /

$$S_1^2 = \frac{\Sigma(x-\bar{x})^2}{n_1-1} = \frac{84.4}{8-1} = 12.0571$$

$$S_2^2 = \frac{\Sigma(y-\bar{y})^2}{n_2-1} = \frac{102.6}{10-1} = 11.4$$

Calculate F- statistic

$$F = \frac{S_1^2}{S_2^2}$$

$$F = \frac{12.0571}{11.4} \approx 1.0576$$

$df_1 = 7$, $df_2 = 9$
$(n_1-1) = 8-1 = 7$
$(n_2-1) = 10-1 = 9$

$1.0576 < 3.29$
Since F statistic is less than the critical value, we fail to reject the null Hypothesis. Ans

Qu-04 What do you mean by SQC?
[AKTU-2021-22]

Sol^n SQC Stand For Statistical Quality Control is a method used to ensure that Products and Services meet Certain quality Standard by using Statistical technique.

Page No.:
Date: / /

Qu-05 Define Null Hypothesis ? [AKTU-2022-23]

Sol^n Null Hypothesis is define that assumes there is no diff^n in a situation or experiment. Tested under the assumption is true. representation is H_0.

Qu-06 Test the effectiveness of incolution against Cholera the Following table was observed. [AKTU-2009, 2018, 2021] χ^2-test ?

	Attacked	Not attacked	Total
Inoculated	30	160	190
Non-Inocu^m	140	460	600
Total	170	620	790

1. Null Hypothesis :-
The Inoculation does not prevent attack From cholera

2 Contingency table

	Attacked	Not Attacked	Total
Inoculated	$\frac{190 \times 170}{790}$ $= 40.84$	$\frac{190 \times 620}{790}$ $= 149.11$	190

Not Inoculated	$\frac{600 \times 170}{790}$ $= 129.11$	$\frac{600 \times 620}{790}$ $= 470.89$	600
Total	170	620	790

χ^2- test

O_i	E_i	$\frac{(O_i - E_i)^2}{E_i}$
30	40.89	2.898
160	149.11	0.795
140	129.11	0.918
460	470.89	0.252

$$\sum \frac{(O_i - E_i)^2}{E_i} = 4.863$$

2 $DOF = (r-1)(s-1)$

$= (2-1)(2-1)$

$= 1 \qquad \chi_{0.05} = 3.841$

$\chi^2 > \chi^2_{0.05}$

H_0 is rejected.
Hence we defend the Statement that Inoculation Prevent attack From Cholera. Ans

Page No.:
Date: / /

Qu-07 From the Following table regarding the Color of eyes of father & son, test if the Color of son's eye is associated that of father

Eye Color of father	Eye-Color of Son: Light	Not Light	
Light	471	51	522
Not-Light	148	230 = 281	378
	619		900

$\chi^2_{0.05(1)} = 3.841$ [AKTU-2020-21]

Soln 1. Null Hypothesis H_0:- The color of son's eye is not associated with that of father i.e they are independent

2 ~~No~~ Contingency table :

eye color of son / eye color of father	Light	Not-Light	Total
Light	$\frac{619 \times 522}{900} = 359.02$	$\frac{522 \times 281}{900} = 162.98$	522
Not Light	$\frac{378 \times 619}{900} = 259.98$	$\frac{378 \times 281}{900} = 118.02$	378
Total	619	218	900

Page No.:
Date / /

3 χ^2- test

O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
471	359.02	111.98	12539.5024	34.92
51	162.98	-111.98	12539.5204	76.93
148	259.98	-111.98	12539.5204	48.23
230	118.02	111.98	12539.5204	106.24
				266.32

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 266.32$$

4. $\chi^2_{0.05}$ for DoF $= (r-1)(s-1)$
$= (2-1)(2-1) = 1$

5 Conclusion :- $\chi^2 > \chi^2_{0.05}$
→ H_0 is rejected. the colour of some's eye is associated with the father.

SUBSCRIBE
JOIN TELEGRAM